30401 Mathematics and Statistics - Module 2 (Statistics) - BEMACS

FIRST PARTIAL EXAM, Version A - 60 minutes

Surname		Name		Student Number	

I commit to respect the Bocconi Honour Code. Signature:

Some exercises refer to the **Bidding** dataframe in **Bidding.rda**. Airlines allow economy passengers to bid for upgrades to business class, with the highest bids receiving the upgrade. This R dataset contains a sample of bids from **a specific demographic segment**. Variable descriptions are provided where needed, and some variables are normalised to allow comparisons across flights.

REFER TO THE SCRIPT FOR ALL CODE NEEDED

FOR THE REPORTED ANSWERS

EXERCISE 1 (8 points, R Dataset and Frequency Table)

For this exercise refer to bids for a business class upgrade of **two samples of customers** from different demographic segments:

(i) one is in the R dataset (Bidding.rda, variable **Bid**),

(ii) the other one is grouped in frequency table below.

Bid Class	[0,20]	(20,50]	(50,60]	(60,70)	(70,100]
Relative Frequencies	0.15	0.30	0.20	0.20	0.15
	0.0075	0.01	0.02	0.02	0.005

a) (1pt*) Calculate and report the median, the first quartile and third quartile of the bidding amount (variable **Bid**) for the sample of customers **in the R dataset**.

b) (1pt*) For each Bid Class **in the frequency table**, report its frequency density.

c) (2pt**) Calculate and report the (approximated) median, the first quartile and third quartile of the submitted bid for the **frequency table**.

d) (2pt*) For each of the two samples (**frequency table** and **R dataset**) calculate and report the (approximated) mean.

e) (2pt*) Using the calculated quantities, compare the shapes of the two distributions.

a) $Q_1 = 39.67$ $\quad Me = 47.96$ $\quad Q_3 = 56.87$

b) $d_i = \frac{p_i}{w_i}$. Frequency densities are reported in the table

c) $Q_1 = 20 + \frac{0.25 - 0.15}{0.01} = 30$ $\quad Q_3 = 60 + \frac{0.75 - 0.65}{0.02} = 65$

$Me = 50 + \frac{0.5 - 0.45}{0.02} = 52.5$

d) Mean – dataset : 47.5

Mean – table : $\sum mid_k p_k = 48.75$

e) dataset : $Me \approx Mean$ and $Q_3 - Me \approx Me - Q_1$

this indicate symmetry which is confirmed from histogram (not perfect symmetry)

table : $Mean < Me$ and $Me - Q_1 > Q_3 - Me$

indication of mild left skewness

EXERCISE 2 (10 points, R Dataset)

We are interested in comparing the distributions of the amount bid for a business class upgrade (**Bid**) by customers who purchased their tickets using different channels (**Channel** variable, with categories: **Aggregator** – purchase through a fare comparison platform; **Agency** – purchase via online agency; **Airline** – purchase through the airline's own website).

a) (2pt*) Provide a sketch of the graph suitable for comparing the distributions. Based on the graph, is there association between the variables **Bid** and **Channel**? Explain why or why not.

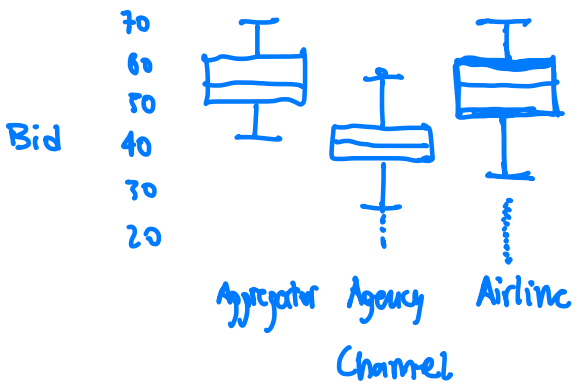
b) (2pt*) Which two summary measures would you use to compare the centres of the distributions considered in the previous point? Compute and report the measures you would use.

c) (2pt*) Considering only the bids submitted by customers who purchased their ticket on a fare comparison platform (**Channel** = **Aggregator**). Provide the five number summary of the variable **Bid** for these Aggregator customers**.** What is the maximum bid among the 5% of customers who bid less?

d) (2pt**) Can we conclude that a bid amount of 35 for the upgrade (**Bid**), submitted by a customer who purchased their ticket on a fare comparison platform (**Channel** = **Aggregator**), is extremely low (outlier)? Show your calculations and all values used to get to your answer.

e) (2pt*) Now consider the relationship between the amount offered by customers for a business class upgrade (**Bid**) and the price they paid for their economy class ticket (**PaidFare**). Discuss the relationship between the two variables including its strength, direction and whether it is a linear relationship. Clearly state which tools you use.

(a)



The multiple boxplot shows the conditional distribution of bid for the three channels. There's clear association between the variables as the amount of bid clearly depends on the channel.

As an example, agency customers seem to bid less with respect to other customers.

(b) Mean and Median (mode is not relevant)

	Aggregator	Agency	Airline
Mean	56.3	40.5	54.3
Median	55.3	40.8	55.5

(c) Five number summary of aggregator customers:

Bid Min: 42.1 Q_1: 50.8 Median: 55.3 Q_3: 62.7 Max: 69.7

Among these customers, the 5% who had the least bid 45.6 or less (5th percentile)

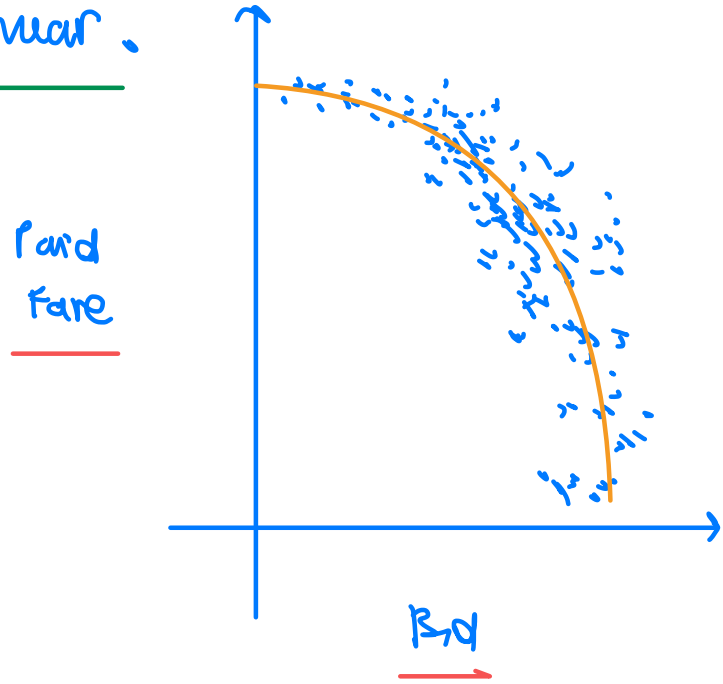
(d) For this group, a lower outlier wrt bid should be below

$Q_1 - 1.5\,IQR = 50.8 - 1.5(62.7 - 50.8) = 32.98$.

Hence 35 cannot be considered a lower outlier.

(e) We check the relationship between two numerical variables with a scatterplot.

The relationship between Bid and Paid fare is strong, negative and non-linear.



EXERCISE 3 (6 points, R Dataset)

The variable **LeadTime** indicates the time in advance the ticket was purchased (**Early** – well in advance; **MediumTerm** – average advance; **LastMinute** – close to the departure date). The variable **Channel** indicates the channel used to purchase the ticket, with the following categories: **Aggregator** – fare comparison platform; **Agency** – online agency; **Airline** – airline website).

a) (2pt*) Identify which type of variable is **LeadTime** and report its distribution through a frequency table with relative frequency.

b) (2pt*) What is the main channel used by passengers who purchase the ticket well in advance? Justify your answer.

c) (2pt**) **Report** and **compare** the two percentages of customers who purchased their ticket with an average advance (**LeadTime** = **MediumTerm**) among those who bought the ticket through a comparison platform (**Channel** = **Aggregator**), and among those who did it through an agency (**Channel** = **Agency**).

(a) LeadTime is a categorical variable (both nominal and ordinal are acceptable).

	Early	Last Minute	Medium Term
Rel. Freq.	0.21	0.41	0.38

(b) Passengers who purchased well in advance, do it mainly through the Airline (51.4% of them), then aggregator (35.7%) and the agency (12.9%)

(c) Among purchases through the aggregator, 50.7% were done medium term, which is twice as much as the percentage of medium term purchases among those done via agency, 25.2%.

EXERCISE 4 (7 points)

An airline operates short-haul European flights with 150 seats. To reduce losses due to cancellations and no-shows, the airline uses overbooking: up to 155 tickets are sold for each flight one week before departure.

Assume that, for a fully booked flight (i.e. 155 tickets sold), the number of passengers who cancel, change their flight, or do not show up is distributed according to a **Poisson random variable with mean parameter 10**.

a) (3pt*) For a flight that is fully booked for 155 seats one week before departure:
   i) What is the probability that more passengers show up than the 150 seats available?
   ii) What is the probability that the flight has at least two free seats?
   iii) How many free seats will have the flights with the 5% least amount of passengers?

b) (2pt**) Consider 200 such fully booked flights.
   i) On average, how many of these flights will have more passengers showing up than available seats (overcapacity)?
   ii) What is the probability that at most 1 of these flights experience overcapacity?

c) (2pt***) Suppose that, whenever a flight is overbooked, each passenger who is denied boarding receives a compensation of 1000 euros.
   i) On average, how much does the airline spend on compensation for one fully booked flight?

      The airline estimates that each additional ticket sold above 150 generates a gain of 50 euros. For example, selling 155 tickets generates an additional revenue of 5·50 = 250 euros, before accounting for compensation costs.

   ii) Let x denote the number of tickets sold in excess of 150. For which values of x is the overbooking strategy profitable, i.e. when does the expected gain $50 \cdot x$ exceed the expected compensation paid to passengers who are denied boarding?

Define $X :=$ passengers who do not show up. $X \sim \text{Poisson}(10)$

a) i) $P(X < 5) = \text{ppois}(4, 10) = 0.0293 =: p$

ii) $P(X \geq 7) = 1 - \text{ppois}(6, 10) = 0.8699$

iii) $\text{qpois}(0.05, 10) = 15$. Hence in the 5% flights with least passengers we have $155 - 15 = 140$ passengers or less.

b) i) $p = 0.0293$ $n = 200$ $p \cdot n = 5.8505$

ii) Poisson modelling $P(Y \leq 1) = \text{ppois}(1, 5.8505) = 0.0197$

$Y \sim \text{Poisson}(np)$

(c) There are $\boxed{m=5}$ cases where the company has to pay.

(1) All people show up $P(X=0) = \text{dpois}(0,10) = 1.53 \cdot 10^{-4}$

5 people to compensate cost: 5000

(2) 1 person does not show up $P(X=1) = \text{dpois}(1,10) = 0.000154$

4 people to compensate cost: 4000

⋮

(5) 4 people doesn't not show up $P(X=4) = \text{dpois}(4,10) = 0.0189$

1 person to compensate cost: 1000

$$\sum_{i=0}^{m-1} P(X=i) \cdot 1000\,(m-i) = \text{sum}(\text{dpois}(0{:}m,10) \cdot (1000 * (m{:}1))) = 12.90$$

$m-1$

$m=5$

More easily done via simulation:

```
N = 1000000
not_showing = rpois(N,mu)
compensation = pmax(0,5-not_showing)*1000
mean(compensation)
```

(ii) Try different value of m in the code above until the expected cost exceeds m·50.

This happens for m=8

Expected cost 460.35

Expected gain 400

So overbooking is profitable up to m=7.